

PROPOSAL

Terrain Generation Using Perlin Noise and Erosion Simulation

Motivation

Generating a realistic terrain plays a crucial role in creating immersive environments for video games and simulations. Realistic terrain enhances the visual appeal of the environment and adds a layer of depth that makes it more engaging for the user. Perlin noise creates smooth, natural-looking terrain, but adding hydraulic erosion helps introduce realistic erosion patterns that can be found in real-world landscapes. This project explores how combining Perlin noise with a hydraulic erosion simulation can create more believable and dynamic terrains.

Description

This project's goal is to implement a terrain generation tool that uses Perlin noise to generate a heightmap for a terrain mesh. The generated terrain will have the option for users to modify parameters, providing flexibility in terrain design. Once the heightmap is generated and applied to the terrain mesh, a hydraulic erosion effect will be applied to the terrain using an erosion technique. This effect will simulate the natural way water would flow on the surface of the terrain, creating more realistic terrain features like valleys and hills. The system will also include an interface that will allow the user to easily generate the terrain, tweak parameters, and apply the erosion effect. This project will be developed in Unity, allowing it to be easily reused for video game development in the future.

Objectives

1. Create a Customizable Mesh with adjustable resolution

This objective is to create a system that allows users to decrease the number of vertices if needed. This will allow for flexibility in the terrain's level of detail and will showcase how the noise affects different levels of terrain complexity.

2. Implement Perlin Noise

This objective focuses on developing the core logic of the Perlin noise algorithm. The generated heightmaps will be applied to the mesh to create varied terrain.

3. Implement erosion Logic

This objective focuses on developing and integrating a hydraulic erosion algorithm that will increase the realism of the terrain by simulating the effects of water flow.

4. Design a User Interface for Terrain Generation and Manipulation

The goal here is to create a simple and intuitive interface that lets users generate a heightmap and mesh, manipulate various Perlin noise parameters, observe the terrain, and apply hydraulic erosion to modify the landscape.

5. Demonstrate Realistic Terrain Features

This objective is to showcase a variety of terrain examples with features like eroded mountains and valleys demonstrating how noise and erosion work together to shape landscapes.